

Instructor: Date:

Learner: Regional Branch:

Program: Center:

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Let's Start!

Help, it's not loading!



Learning Outcome:

Interpret the technical design report, to determine the work plan during the phase of building the software, based on regulation and protocols established by the company.

Design the architecture design using the appropriate tools and types according to the software requirements.

Ask your partner:

- What are the possible reasons for software fails?
- What possible solutions are there to fix software?

Can Do Statements:

- I can describe the most common architectural designs in charge of communication.
- I can talk about the most common architectural designs that focus on deployment.
- I can give information about the most common architectural designs according to their domain and structure.





Let's Explore!

Martin reading an article about software architectures.



Instructions: Read the text below.

Just as a house, a program needs to be planned and designed before being built so that it doesn't collapse. Such a process is what we identify as software architecture. It is in charge of providing a combination of lexicon of components and connectors as well as solutions to possible glitches or complications, guaranteeing this way the best functioning of the system. Now, depending on the purpose of your software you might want to analyze the different architectural styles that are there.

Depending on their category, the architectural designs might change. For example, there are 4 main categories: communication, deployment, domain, and structure. Within the first category we can find two of the most common architectural styles, the message bus and the service oriented architecture (SOA). You ought to make use of the first one to receive and send messages by means of more than one communication channel. Similarly, you might want to exploit SOA to define the communication methods, in terms of software that you might want to use in your company.

Once you've decided what communication method to use, you'd better choose one deployment method as well, to do so, you ought to choose between one of the most used ones, that is, the client/server method and the 3-tier one. Either way, you must analyze the best option for your company, as the client/server architecture will split the communication into two, allowing the server to communicate directly with the client and vice versa. On the other hand, the 3-tier one will divide the deployment of communication into three segments, all in different stations.

Finally, you might want to choose the best architecture regarding the domain and the structure of your software. The domain driven design and the component based architectures are two of the best architectures you might want to implement, as they both focus on communication channels within the domain. Similarly, the layered and object oriented architectures ought to be analyzed, as they outline the best way to manage the software, in terms of structure; here you'd better select the one that best divides the software, that is, layers or objects that contain the data and behavior of your system.





Let's Practice!

Discussing Domain and Structure Architectural Designs

Instructions: Select the correct answer.

1. A component based architecture provides one of the _____ communication interfaces.
 - a. best ways to define
 - b. easiest problematics
 - c. best possibilities to
 - d. worst ways of communicating
2. The _____ architecture is the Layered architecture.
 - a. best way to define
 - b. most common design
 - c. most regular architecture
 - d. best design
3. They had better apply an object oriented design. It is _____ for dividing element responsibilities.
 - a. most common
 - b. the most bought
 - c. most applied one
 - d. the most applicable one
4. In terms of structure, you'd better select the one _____ the software, that is, layers or objects.
 - a. that bests
 - b. the best divided
 - c. that best divides
 - d. the best division of
5. You should redistribute your code by using an object oriented design architecture. It is _____ for your needs.
 - a. the most suitable
 - b. the best suitable
 - c. the most
 - d. the best



Let's Practice!

They talk about the software security system.

Instructions: Match column A with Column B.

1. the / the / most / What / architecture? / characteristic / important / is / of / layer

2. into / layout / Object / architectures / a / of / program / the / oriented / divides / objects.

3. application / architecture / layered / called / into / sets / The / the / groups / layers.





Extension Activity

It's time for you to show how much you've learnt about software architectural designs.

Instructions: Complete the following steps. Fill in the Self-Assessment.

- Research about the different types of software architectural designs.
- Select three types of architectural designs and make a brochure comparing and describing their uses, pros and cons.
- Share your brochure with your classmates and exchange.

Self-Assessment

Yes

No

- I read about the different types of software architectural designs.
- I make a brochure about the software architectural designs.
- I share my findings with my classmates and instructor.

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Target Words

- **lexicon** (noun): specific or technical vocabulary
e.g. It is in charge of providing a combination of lexicon of components and connectors.
- **glitch** (noun): problem or complication
e.g. It is in charge of providing a combination of lexicon of components and connectors as well as solutions to possible glitches or complications.
- **split** (verb): to separate into two or more parts
e.g. Either way, you must analyze the best option for your company, as the client/server architecture will split the communication into two.
- **station** (noun): a computer or a server located in an isolated place
e.g. On the other hand, the 3-tier one will divide the deployment of communication into three segments, all in different stations.
- **outline** (verb): to design or draft something
e.g. the layered and object oriented architectures ought to be analyzed, as they outline the best way to manage the software, in terms of structure.



Learning Points

Ought to

You can express ideas that show advisability by using “**ought to**.”

Ought to is a modal verb; as a modal verb, it follows the next rules:

- Modal verbs always come after the subject and before the main verb.
- The main verb is always in infinitive form, regardless of the subject (I, you, he, we, etc.).
- A modal verb should never follow another modal verb in a sentence.

In terms of meaning, “**ought to**” is considered to have the same meaning as “should” and similar to “had better” when it is used for giving advice.

In order to make a sentence with ought to, you can use the following structure:

Subject + ought to + infinitive verb + complement.

Example:

You ought to identify what the needs of your software are.

Notice that the use of “ought to” in interrogative form is not common, therefore people rather use should instead of ought to when asking questions.

Examples:

~~Ought (to) you choose a better software architecture?~~ (wrong)

Should you choose a better software architecture? (right)

Examples

- *You ought to make use of the first one to receive and send messages by means of more than one communication channel.*
- *You ought to choose between one of the most used ones, that is, the client/server method and the 3-tier one.*
- *Similarly, the layered and object oriented architectures ought to be analyzed, as they outline the best way to manage the software.*



Learning Points

Common Architectural Designs

Architectural designs provide the software with the framework and patterns to work with. They provide rules for combining and configuring the different components as well as the ways to communicate with each other.

Depending on their area of focus, architectural designs can be organized into 4 basic categories: communication, deployment, domain and structure.

Compare:

Category	Architectural Design	Brief description
Communication	Message bus	Indicates the use of software that sends and receives messages using more than one channel.
	Service-Oriented Architecture (SOA)	Allows the program to send and receive several messages or services to a server through different terminals.
Deployment	Client/server	It is a direct connection between the client and the server.
	N-tier (3-tier)	Divides the tiers into three, where the client does not communicate directly to the server but to a sub server to process more information.
Domain	Domain Driven Design	Implements the business model to link the components in the program.
	Component based	It decomposes the components of the software to individual components.
Structure	Layered	It layers the functions of the components so that a layer can be modified without affecting the others; it is only linked to the previous layer
	Object oriented	It divides the components into independent and reusable objects.